



NU LAGUNA

Network and Communications

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Router Interfaces Configuration

LESSON OUTLINE

- Router Interface Configuration
- Fast Ethernet Configuration

LEARNING OUTCOMES

Here's what I will teach you in this course material:

- Apply the Class C and Class B addressing when allocating IP Address in the devices.
- Perform interface configuration tasks on a router.
- Use the show commands.

RESOURCES NEEDED

For this lesson, you would need the following resources:

- Packet Tracer

STARTING POINT

Pretest

Before you start, try answering the following questions. Determine the name of the port in the image below.







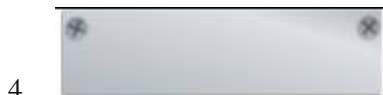




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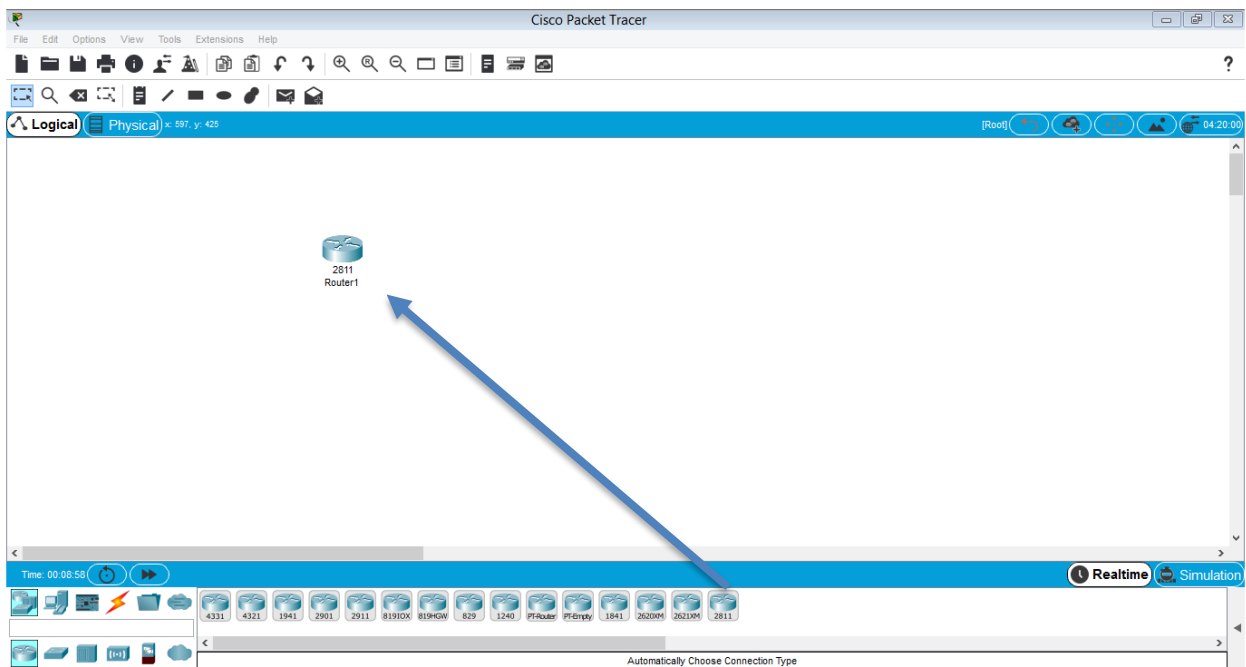
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References

Router Interface Configuration

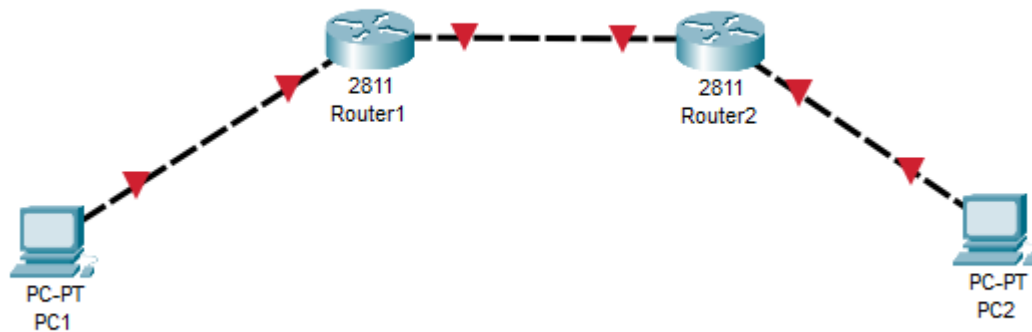
By this time, you should already have the background in basic router configuration. Open your Packet Tracer and select the router from the lower left-hand corner and drag it into the center of the sandbox screen as seen below.



You will be setting up a basic network, so the next step is to select devices from the bottom left-hand corner and drag it to the sandbox screen. Do the same illustration shown below.

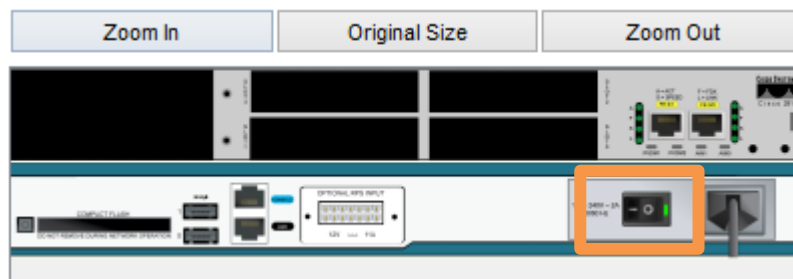


Now select connections from the same bottom left-hand corner. When you connect like-devices (Such as a router and computer) you use a crossover cable, so you should choose a copper crossover cable from the second menu to the immediate right. You may notice that even though a link is established, it is not functional. You will be able to know by the red dots that are present on both ends of the connection. Once the router is configured correctly, the red dots will become green to imply the devices can communicate. Your connections must be the same as the image shown.



Fast Ethernet Configuration

The router that is turned off does not work very well! Click on your router1 and router2 to bring up the configuration menu and verify that it is turned on. When on, there will be a small green light below the switch as seen in the diagram.



Next, you must open the Ethernet ports to enable communication. Though they are physically connected, they are in a state that is known as being in administrative shut down.

Configuration of Interfaces

For Fast Ethernet

Syntax:

Router(config)#interface Fanumber

Router(config-if)#ip address hostaddress subnetmask

Router(config-if)#no shutdown

Example

Router(config)#interface Fa0/0

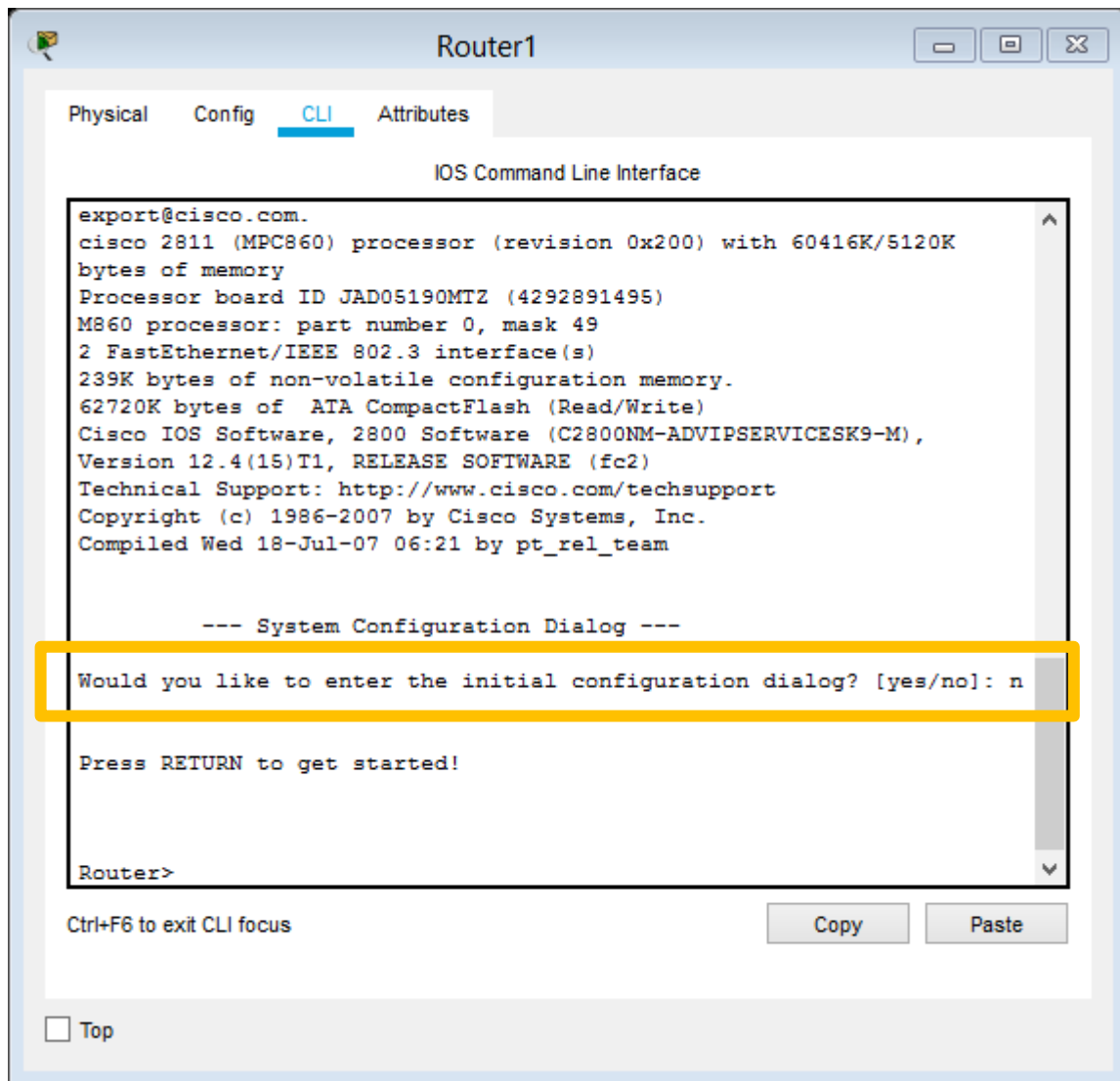
Router(config-if)#ip address 192.168.1.1 255.255.255.0

Router(config-if)#no shutdown

To start your configuration, I will give you the IP addresses you will be using to configure the devices in network topology.

Device	Interface	Host Address	Subnet mask	Default Gateway
Router1	Fa0/0	193.190.30.1	255.255.255.0	N/A
	Fa0/1	191.10.0.1	255.255.0.0	N/A
Router2	Fa0/1	130.60.0.1	255.255.0.0	N/A
	Fa0/0	191.10.0.2	255.255.0.0	N/A
PC1		193.190.30.2	255.255.255.0	193.190.30.1
PC2		130.60.0.2	255.255.0.0	130.60.0.1

Now click on the Router1 CLI tab to access the configuration menu, you will start to configure your Ethernet interfaces. Type n or no in the question “Would you like to enter initial configuration dialog?” as shown in the image.



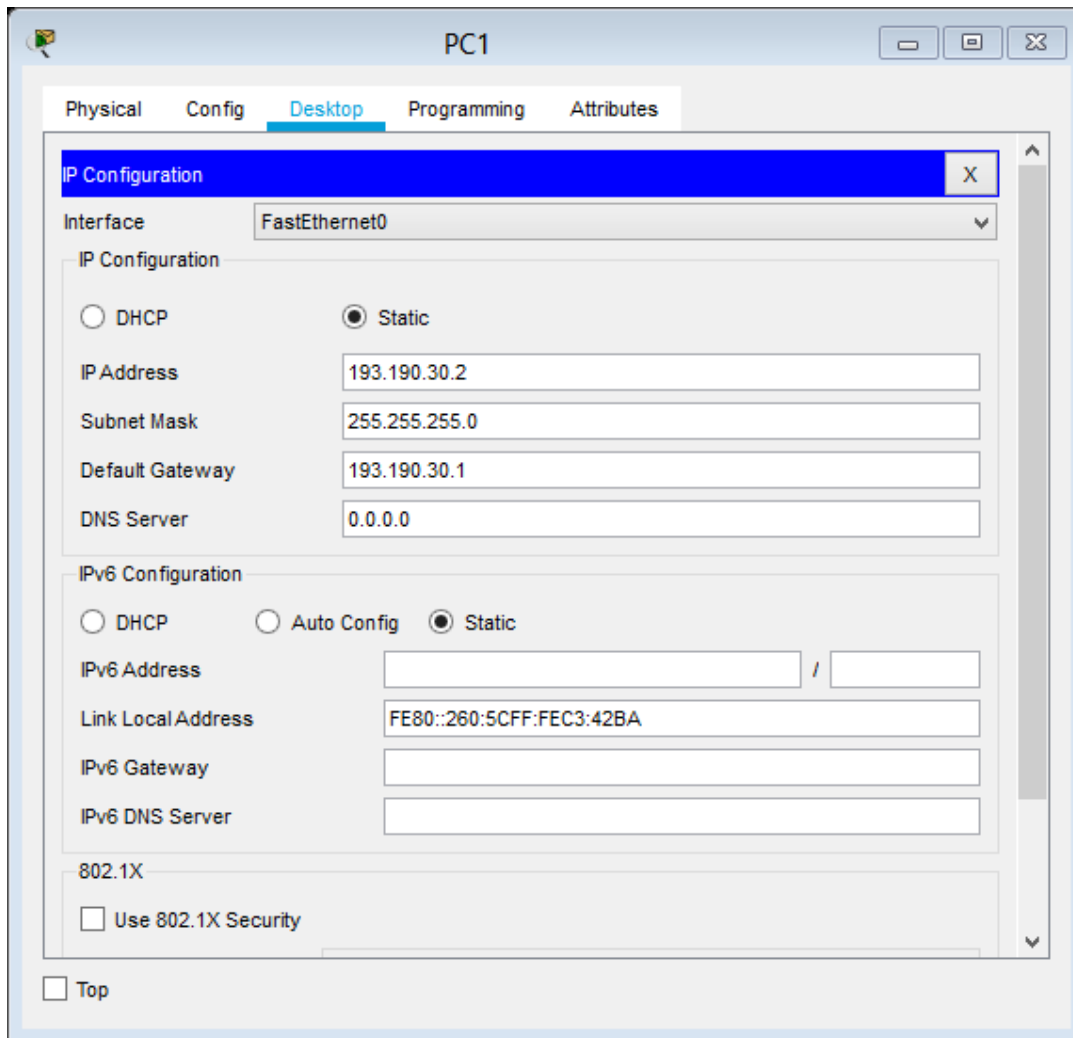
For your Fa0/0 interface, do the code shown below.

```
Router>ena  
Router#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#int fa0/0  
Router(config-if)#ip add 193.190.30.1 255.255.255.0  
Router(config-if)#no shut
```

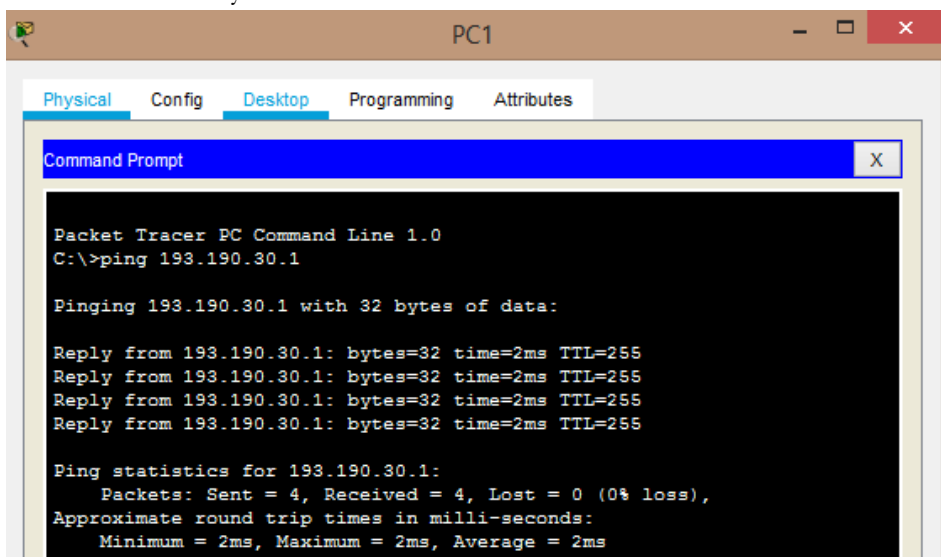
That's it! You should now see a message like the following:

```
Router(config-if)#  
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,  
changed state to up
```

Now you must do the same thing for fastethernet0/1. If you don't, there still won't be a connection to PC1! Make sure to enter the IP address carefully as seen below:



Test the connectivity of PC1 to Router1.



Next step is configuring your Router1 to Router2 interfaces. Go to your Router1 CLI tab then do the following:

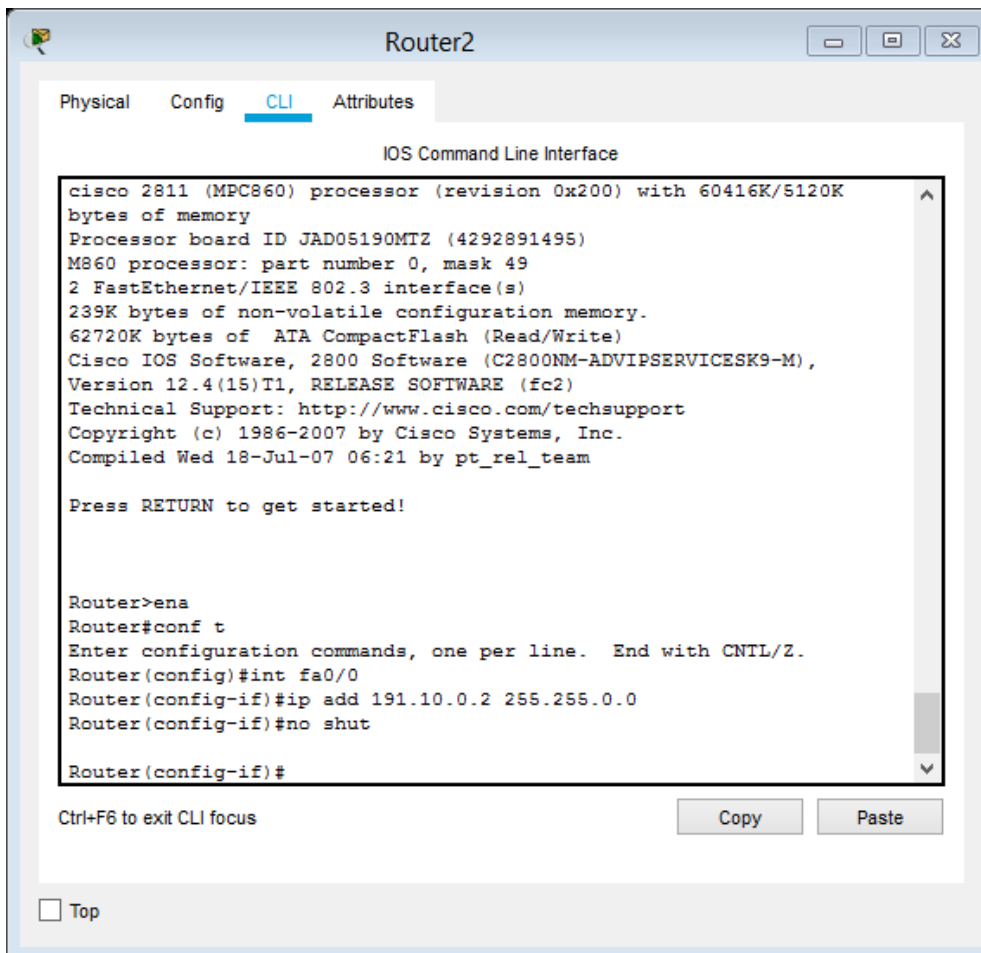
```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip add 191.10.0.1 255.255.0.0
Router(config-if)#no shut
```

Save your configuration to NVRAM.

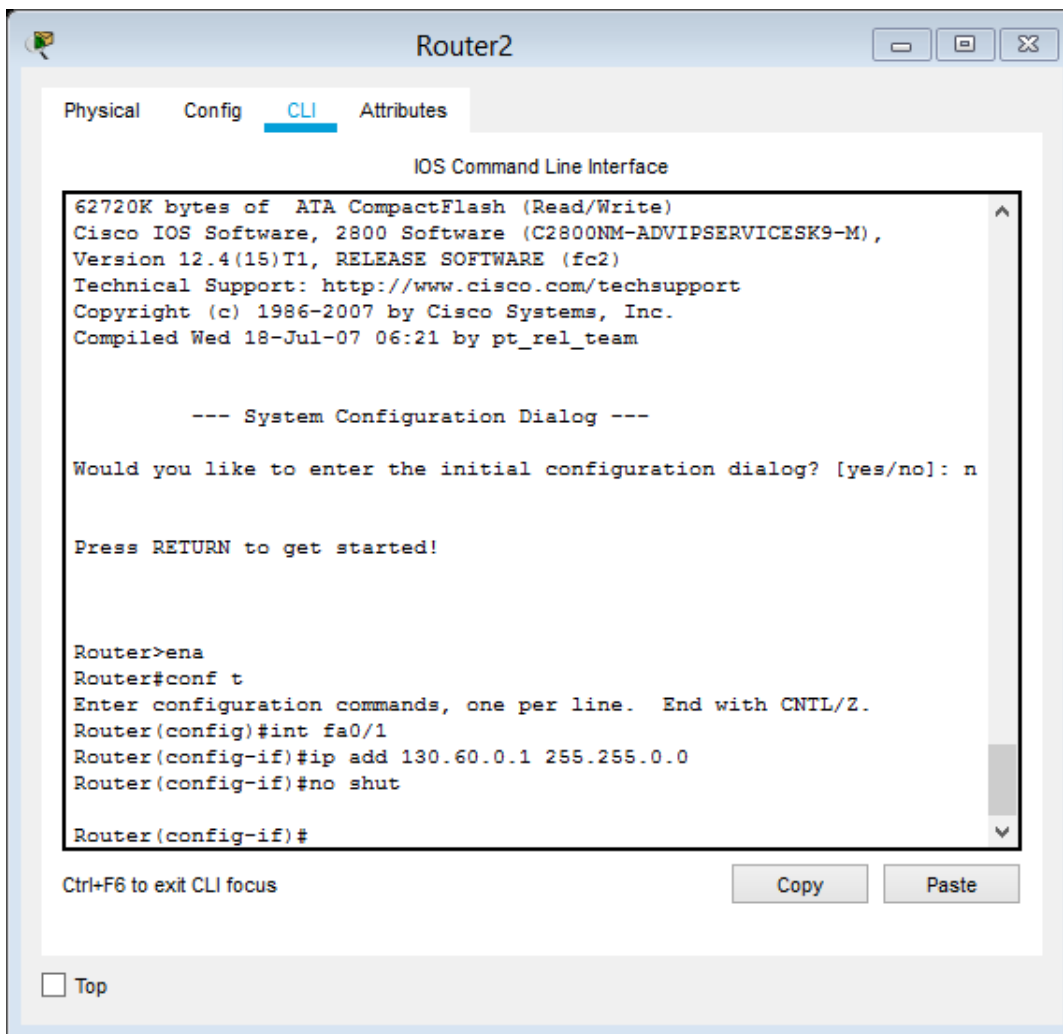
```
Router(config)#exit
Router#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

Configure your Router2 interfaces.

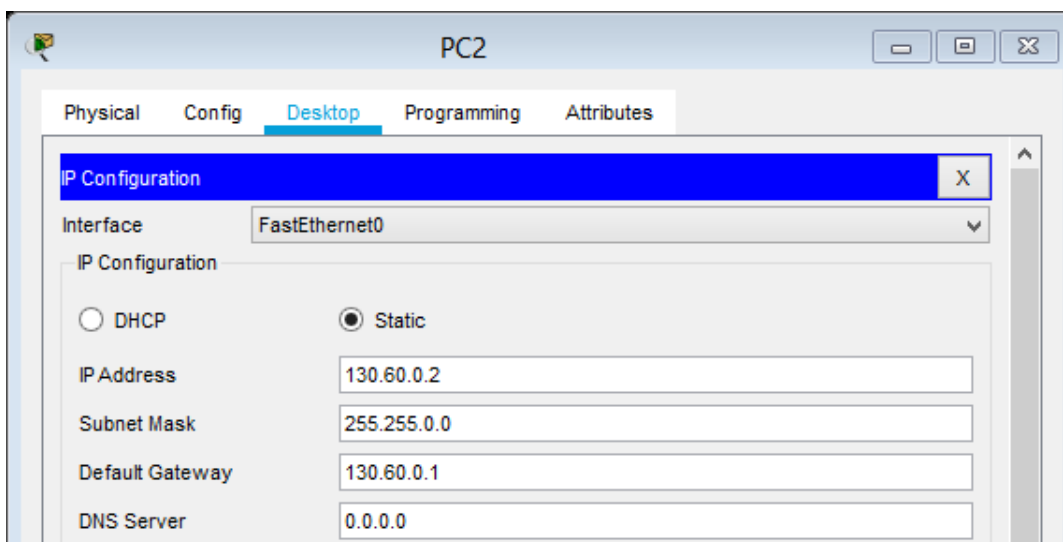
For Fa0/0



For Fa0/1



For PC2



Save your Router2 configuration to NVRAM.

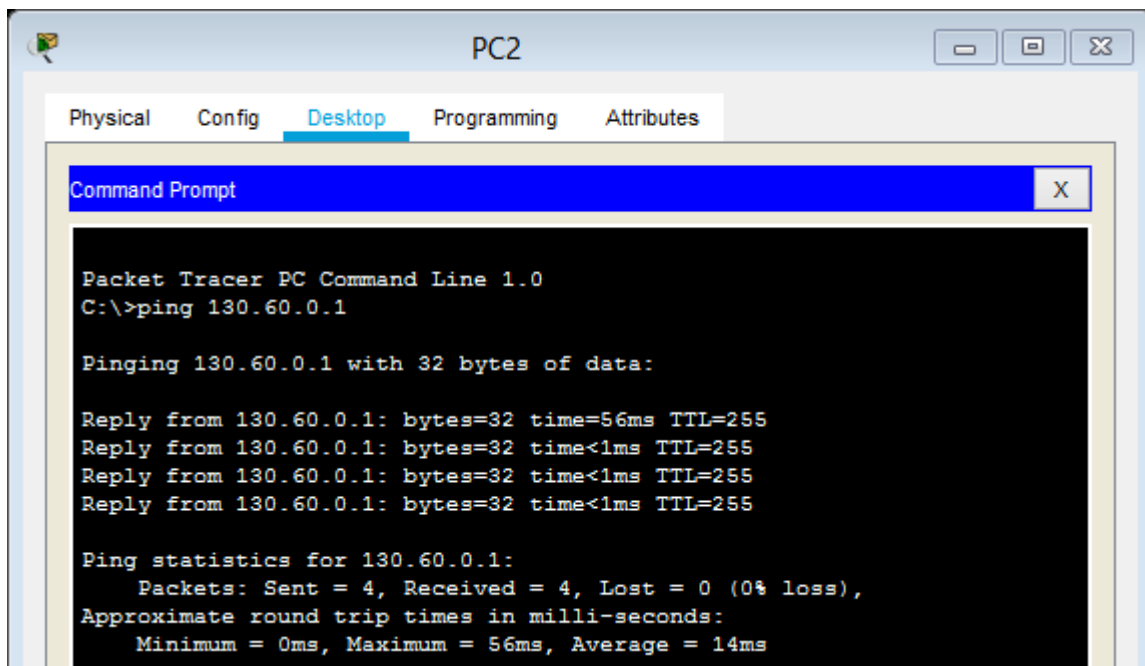
```
Router(config-if)#exit
Router(config)#exit
Router#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

Test your Router2 connectivity. In your Router2 CLI tab ping the IP address of your Router1 interface.

```
Router#ping 191.10.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 191.10.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

In your PC2, ping the Router2 fast ethernet.



Removing assigned IP Address

Syntax:

```
Router(config)#interface Fanumber
```

```
Router(config-if)# no ip address
```

```
Router(config-if)#no shutdown
```

Example

```
Router(config)#interface Fa0/0
```

```
Router(config-if)#no ip address
```

```
Router(config-if)#no shutdown
```

REFERENCES

CISCO Networking Academy. (2016). *Introduction to networks Companion Guide v5.1 (1st ed.)*. USA: CISCO Press

Cisco Networking Academy Program CCNA 1 and 2 Lab Companion (Revised 3rd Edition). Cisco Systems Inc.: Cisco Press.

Cisco Networking Academy CCNA1 Module. Cisco Systems Inc.: Cisco Press.